The Media Connectivity Association



OVERVIEW AND UPDATES · IMMERSIVE MEDIA

Volumetric Video Experiences

Volumetric video platform for immersive experiences

MPEG (ISO/IEC)

OCTOBER 2026 – www.5g-mag.com

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About the project

V3C represents 3D content so that conventional 2D video codecs do the compression

ISO/IEC 23090-5

MPEG standard

Visual volumetric video-based coding (V3C) and video-based point cloud compression (V-PCC).

ISO/IEC 23090-12

MPEG standard

MPEG Immersive Video (MIV), an extension of the V3C base.

ISO/IEC 23090-10

With 3GPP TS 26.512

Carriage of V3C data in ISOBMFF with support for DASH, and the protocols and APIs that deliver V3C DASH presentations over 5G Media Streaming.

FIND OUT MORE **Technical Analysis** **Specifications** **Reference Tools**

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Profiles and components

What V3C defines, and what a V3C bitstream carries

V-PCC ISO/IEC 23090-5: a dynamic point cloud, typically one captured object or performer

- Projection onto per-normal planes
- Packing of projections
- Geometry, occupancy and attribute videos

MIV ISO/IEC 23090-12: several cameras with depth, six degrees of freedom in a limited viewing volume

- Main
- Extended
- Extended Restricted Geometry
- Geometry Absent

Bitstream components clearly separated

- Atlas sub-bitstream
- Common atlas sub-bitstream (MIV)
- Occupancy video
- Geometry video
- Attribute video(s)

Carriage ISO/IEC 23090-10

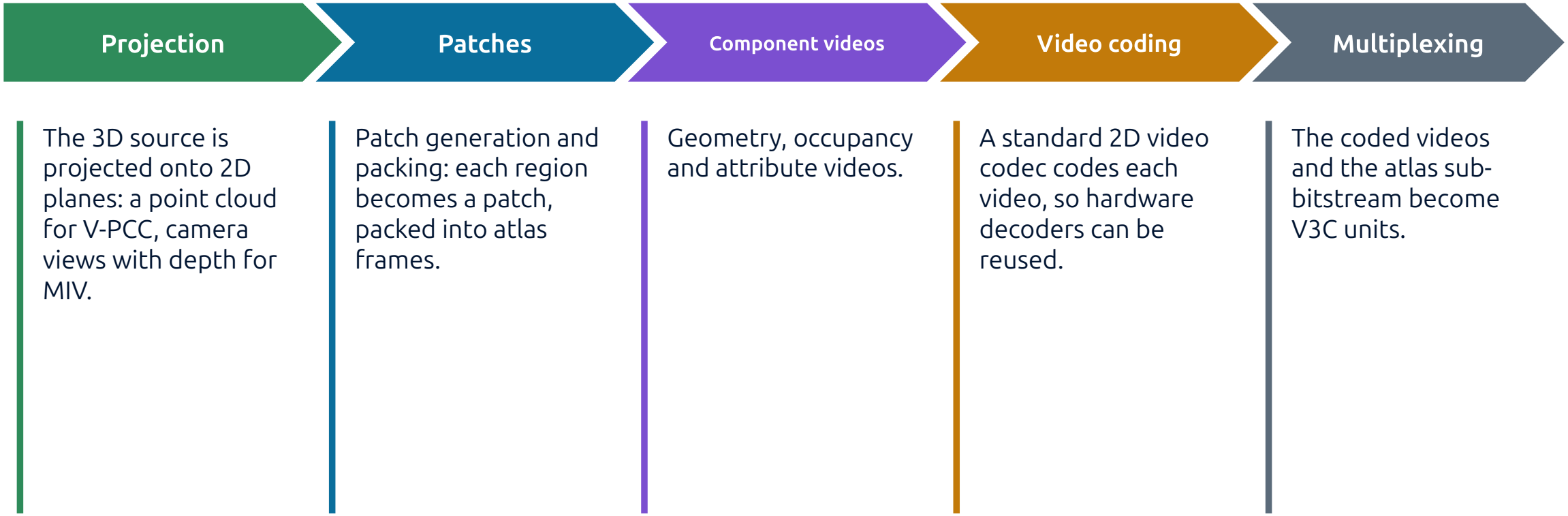
- ISOBMFF
- DASH

The component videos are ordinary coded video, so a decoder can offload them to a hardware video decoder and keep CPU and GPU work for the atlas and the 3D reconstruction.

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How it works

The projection-based pipeline of V3C, from the 3D source to the bitstream



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How it grew

The editions of the MPEG specifications behind the platform

2021	2022	2023	2025	Next
ISO/IEC 23090-5: an edition of V3C and V-PCC is published.	ISO/IEC 23090-10 is first published; an amendment adds support for packed video data.	A new edition of ISO/IEC 23090-5, and the publication of MIV, ISO/IEC 23090-12.	The edition of ISO/IEC 23090-5 that 5G-MAG currently references; later editions support VVC.	A second edition of MIV is under development in MPEG.

Confirm the exact edition and any amendments in the ISO catalogue for the feature set you rely on.

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Repositories

What each repository covers, from its README; support of editions and profiles is not yet audited

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-v3c-unity-player	a Unity package to decode and play V3C content, with a simple demo project	Player · Unity · Windows · Android	● v1.1 30 Sep 2025	GitHub
rt-v3c-decoder-plugin	entry points for decoding and viewing MIV and V-PCC content	Decoder · DASH client · Synthesizers	● v1.1 30 Sep 2025	GitHub
rt-v3c-content	test content for the Unity player, and tools to generate V3C bitstreams and DASH segments	Test content · DASH packager	● v1.1 9 Oct 2025	GitHub
rt-v3c-examples	example V3C content and helper files for testing the encoding, delivery and rendering pipeline	Test content	○ none yet	GitHub
rt-media-origin	a config-driven media origin that packages DASH, HLS or CMAF and delivers over HTTP or push protocols	Media origin · DASH · CMAF	○ none yet	GitHub

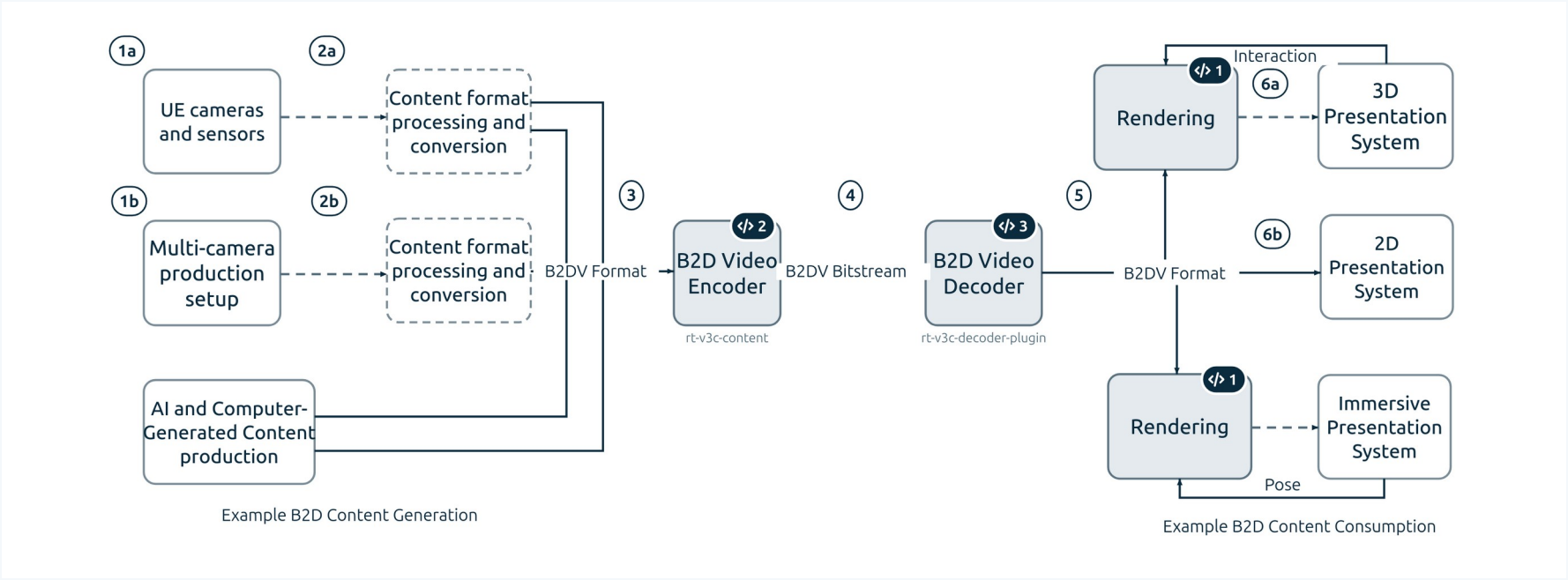
FIND OUT MORE — Technical Analysis · Specifications · Reference Tools

● released ○ no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

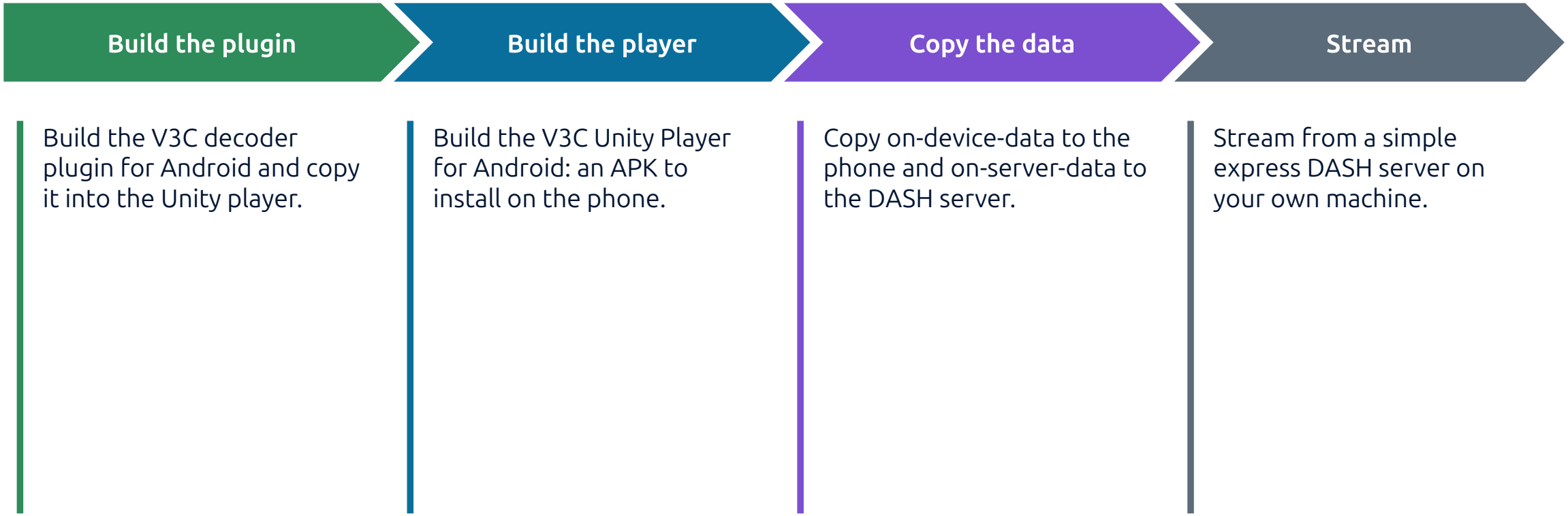
1. **rt-v3c-unity-player**
2. **rt-v3c-content**
3. **rt-v3c-decoder-plugin**
   used by 1

Boxes without a badge are not mapped to a repository.

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Build and try

The tutorial: the V3C Unity Player on an Android phone, streaming from a DASH server



The player needs OpenGL ES 3.2 and hardware HEVC decoding on the phone.

CONTRIBUTE

Latest updates and roadmap

Status on 22 October 2026

RELEASED

rt-v3c-unity-player v1.1

30 Sep 2025. Improves V-PCC rendering, with renamed and extended materials and shaders.

rt-v3c-decoder-plugin v1.1

30 Sep 2025. Adds the source code of the haptic decoder and synthesizer and of the V-PCC synthesizer.

rt-v3c-content v1.1

9 Oct 2025. Adds a V3C DASH packager and tools to generate V-PCC bitstreams.

IN DEVELOPMENT

V3C Unity Player

A fuller player on Android: title and source on screen, content library and settings, playback statistics from the decoder.

V3C Unity Player

Support for new devices: Meta Quest headsets.

Content and origin

Titles that carry their own name, description and environment; real-time push transports in the media origin.

PLANNED

Haptics over DASH

Streaming of haptics over DASH.

Real-time protocols

Support for real-time protocols such as RTP and WebRTC.

More synthesizers

Support for an additional synthesizer, such as V-DMC.

MERGED SINCE V1.0 3 pull requests in 3 repositories

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